

Vivek Vishal

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Summary

Software Engineer (IC4) with 3+ years of full-time experience (4+ years including internship) building scalable, data-intensive systems across mobility, payments, and healthcare. Specializes in distributed systems, AI agent orchestration, real-time data pipelines, and backend architecture. Track record of delivering high-impact features at scale across YC-backed and high-growth startups.

Skills

Languages: Java, C++, Rust, Go, Python, TypeScript, SQL, C

Frontend: React, React Native, TypeScript, JavaScript, Webpack

Backend: Node.js, Express.js, Spring Boot, Django, REST APIs, Distributed Systems

AI / Agents: LLM Integration, ReAct Agent Loops, LangGraph, Pipecat, Prompt Engineering, Tool Calling, Skill Orchestration, Structured Outputs, Observability

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, ClickHouse, NoSQL

Cloud, DevOps & Observability: AWS, Docker, Kubernetes, Kafka, Temporal, Grafana, Prometheus, Kibana, Structured Logging, Alerting

Professional Experience

Asha Health (YC W24), India

Software Engineer (IC4) • Feb 2026 – Sept 2026

- Architected a ReAct agent loop (**ehrCopilotReActLoop**) automating clinic operations end-to-end — from patient scheduling to billing — dynamically loading scoped skills, calling tools against live practice/patient data, enforcing safety-gate checkpoints (e.g. **challenge_my_reasoning**), and logging full execution traces to **CopilotExecutionLog** for replay and RCA.
- Designed a dual-runtime dispatch layer (LangGraph + Asha Native) with lane-based routing across EOB posting, claim generation, pre-auth, claim rejection, and chat workflows; added a forced reasoning-challenge step to the EOB graph, improving output reliability.
- Built and scaled the voice AI agent server for clinics using the Pipecat pipeline and a node-based conversation flow engine, handling **1,000+ customers/day** for real-time front-desk automation.
- Built a Revenue & Analytics service using a CQRS (Command Query Responsibility Segregation) design pattern, decoupling write-heavy billing operations from read-heavy reporting/analytics for improved scalability.
- Improved system observability through structured metrics, logging, and alerting pipelines, ensuring high availability and faster incident resolution across agent runtimes.

Namma Yatri (Juspay), India

Software Development Engineer • Feb 2024 – Jan 2026

- Developed and deployed an ETA prediction system using historical + real-time GPS data, achieving **80% accuracy within 5 minutes**.
- Streamed high-volume telemetry data to Kafka and built real-time analytics pipelines using ClickHouse, enabling sub-second operational dashboards.
- Built route deviation detection and alerting system, reducing incident response time by **30%**.
- Delivered “Where is My Bus” platform integrating GPS, Redis, and alert infra, improving operational efficiency by **25%**.
- Integrated metro ticketing into Google Maps, enabling **5,000+ monthly bookings** across multiple cities.
- Reduced Maps API cost by **Rs.50K/day** via caching and query optimization (cost per ride: Rs.3 → Rs.2.5).
- Launched rental and intercity ride flows, increasing completed rides by **5%**.
- Automated refunds and cancellations pipeline, reducing support ticket volume by **10%**.

Juspay, India

Software Engineer • May 2023 – Feb 2024

- Designed KV infrastructure handling **3K peak RPS** for internal services.
- Developed internal DSL for workflow standardization, reducing developer effort by **66%**.

Juspay, India

Software Engineer Intern • Mar 2022 – Apr 2023

- Built merchant onboarding automation system, cutting onboarding time from **4 weeks → 2 weeks**.
- Built analytics dashboards for merchant success teams to track KPIs and detect issues proactively.

Projects

Multi-Objective Surveillance System

AI-driven object tracking system using YOLO, DeepSort, and Kalman filtering; achieved **50% accuracy** in research simulations.

Humanitarian OpenStreetMap

Contributed to open-source geospatial mapping using Flask and React, supporting community-driven humanitarian initiatives.

Education

National Institute Of Science And Technology, Bhubaneswar, India • Aug 2019 – May 2023

B.Tech in Information Technology | CGPA: 8.2

Coursework: Data Structures, Algorithms, Software Engineering, Machine Learning, Compiler Design